

STAT 221- FUNDAMENTALS STATISTICAL DATA ANALYSIS

Week-1: Introduction;
Data and social science; Science and
Social Science

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BASIC INFO

Course Code - **STAT221**

Course Name - **Fundamentals of Statistical Data Analysis**

Section - **UGSEM3**

OBJECTIVES

To enable with **conceptual understanding about the questions in social science which could be addressed using quantitative methods**

To acquaint **students with data analysis skills to analyse and visualize data** and make meaningful inferences.

To introduce students to **probability, descriptive statistics, and statistical inference.**

To introduce students to **survey and sampling** to address questions of social, economic, political, and public policy issues

COURSE TOPICS

First Half

- Concepts, Variables, and Measurement
- Representation of Data and Data visualization
- Understanding world through data and awareness of biases
- Frequency Tables
- Measures of Central Tendency
- Measures of Dispersion
- Qualitative Data Analysis

Second Half

- Probability- Concepts. Important Probability Distribution
- Sampling and Sample survey
- Inferential statistics- Hypothesis
- Cross-tabulation and Chi-Square
- Measures of Association for Categorical Variables
- Analysis of Variance
- Correlation and Regression

Software tools: **Excel**, visualization softwares- **datawrapper**, rawgraphs, flourish studio

QUANTITATIVE SOCIAL SCIENCE

Data in social science

DATA IN SOCIAL SCI

Family

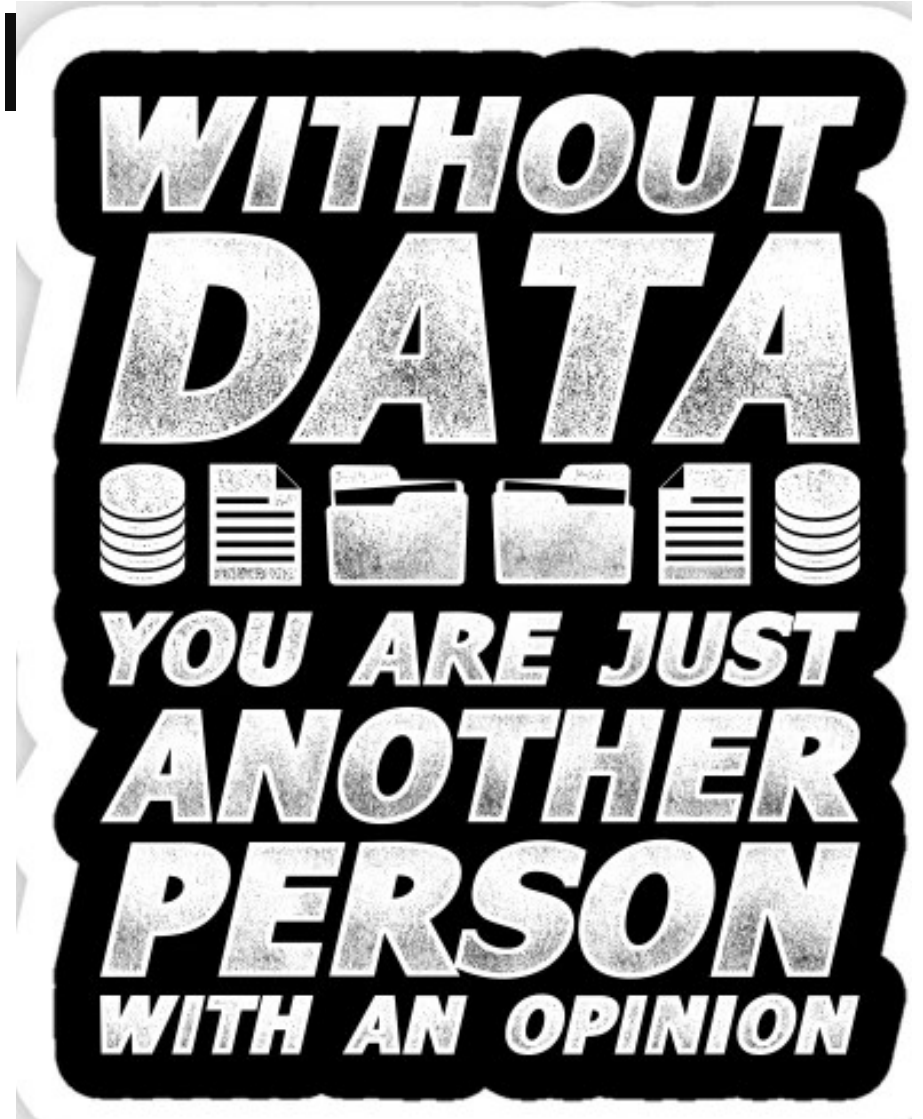
School/College

Society

Administrative unit (geographical):
City, District, State , Nation

Economy

Politics



DATA



SORTED



ARRANGED



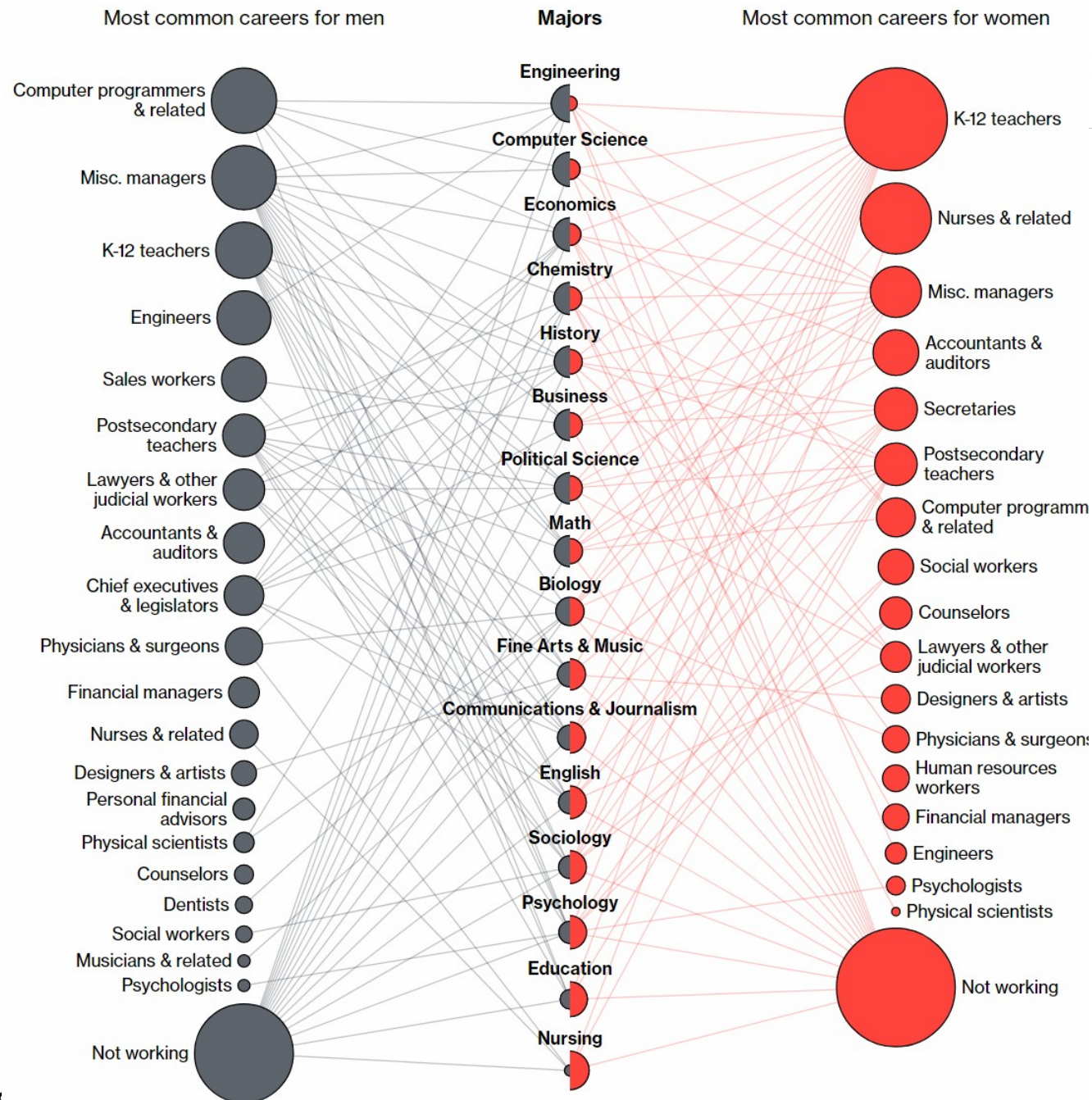
PRESENTED VISUALLY



EXPLAINED WITH A STORY



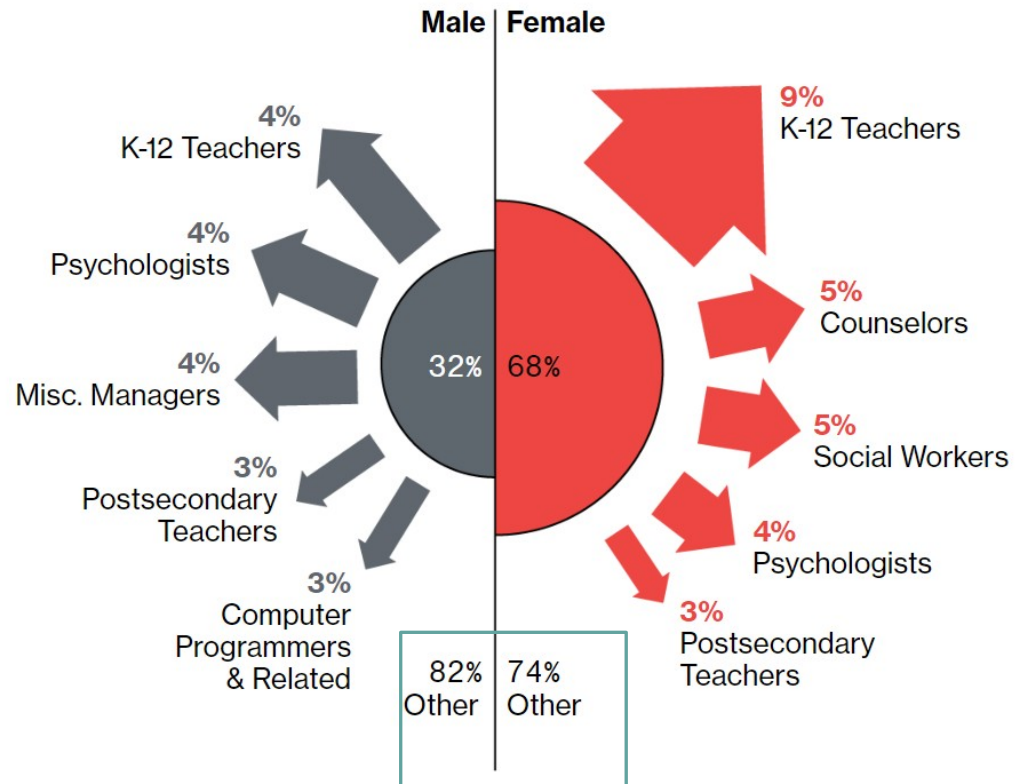
ACTIONABLE (USEFUL)



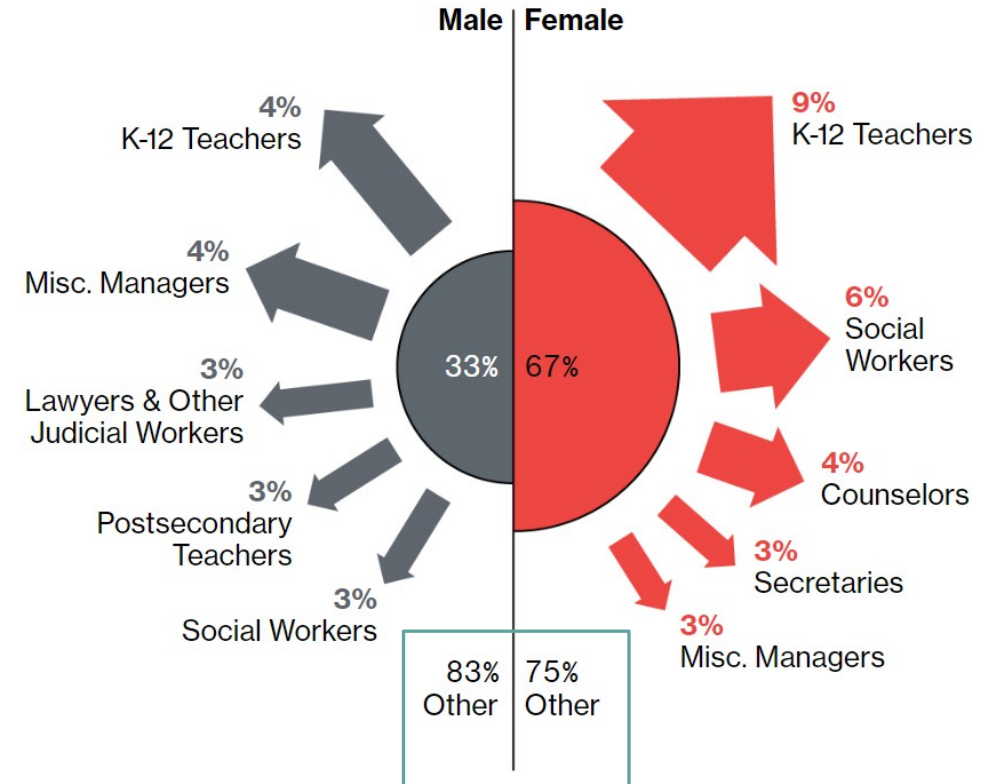
<https://web.stanford.edu/group/sdgc/cgi-bin/yci/p=4927>

<https://www.bloomberg.com/graphics/2018-women-professional-inequality-college/>

Psychology



Sociology



<https://www.bloomberg.com/graphics/2018-women-professional-inequality-college/>



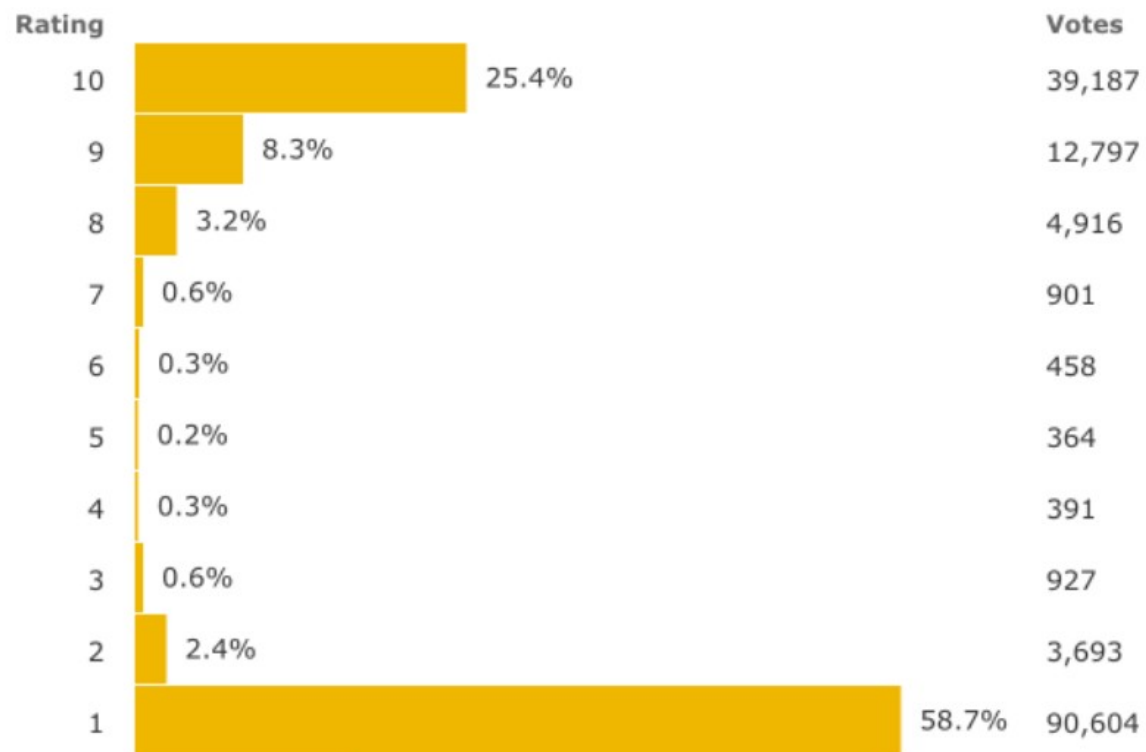
Laal Singh Chaddha (2022)

User Ratings

★ 5 ☆ Rate

IMDb Users

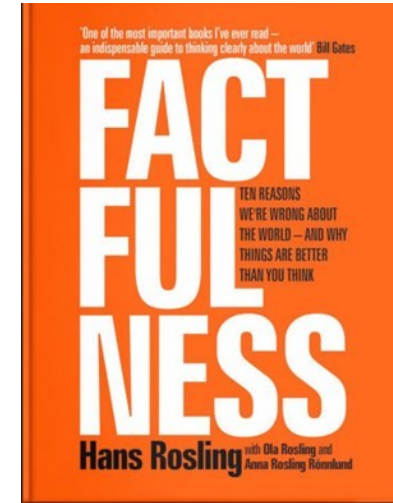
154,238 IMDb users have given a [weighted average](#) vote of 5.0 / 10



Arithmetic mean = 4.3 Median = 1

Understanding Biases in Data

BOOKS /REFERE



Krieg, E. J. (2020). *Statistics and data analysis for social science* (2nd ed.). SAGE Publications.

Hesse, C. A., & Ofosu, J. B. (2017). *Statistical methods for the social sciences*. Akrong Publications Limited.

Rosling, H., Rosling, O., & Rosling Rönnlund, A. (2018). *Factfulness: Ten reasons we're wrong about the world—and why things are better than you think*. Flatiron Books.

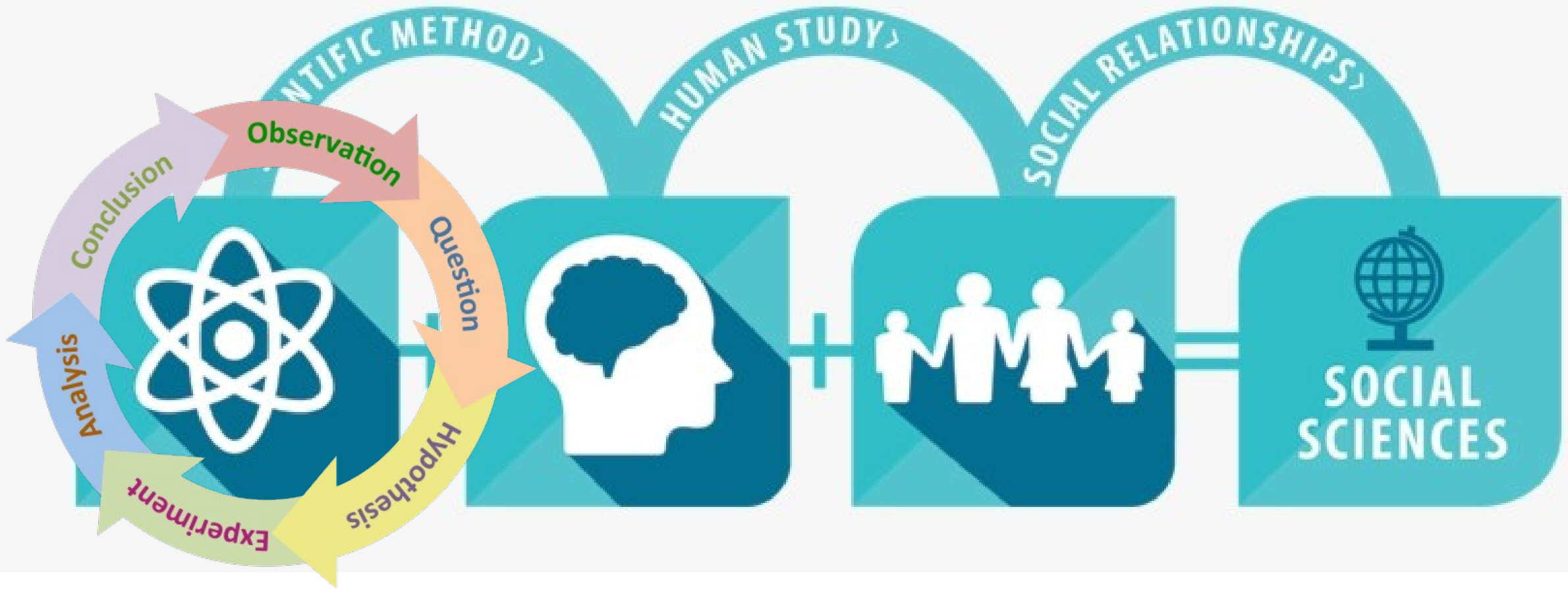
De Veaux, R. D., Velleman, P. F., & Bock, D. E. (2009). *Intro stats* (3rd ed.). Addison-Wesley.

Triola, M. F. (2009). *Elementary statistics* (11th ed.). Addison-Wesley.

Tufte, E. R. (2001). *The visual display of quantitative information* (2nd ed.). Graphics Press.

Tufte, E. R. (1990). *Envisioning information*. Graphics Press.

LECTURE 2: SCIENCE AND SCIENTIFIC METHOD



SCIENCE AND SCIENTIFIC METHOD IN SOCIAL SCIENCES

Image: https://www.pinterest.com/pin/cu2e6e5e0i1q8e65_scientific-method-human-study-social-relationships-science

India approved its own Covid-19 vaccine before it completed its human trials

QUARTZ INDIA

DCGI's Covaxin 'approval' is political jumla. It reinforces idea of Modi's Atmanirbhar Bharat

The Modi govt must not damage India's competitive advantage in vaccines & pharmaceuticals through absurd benchmarks that will not be trusted by other countries.

NITIN PAI 5 January, 2021 8:32 am IST

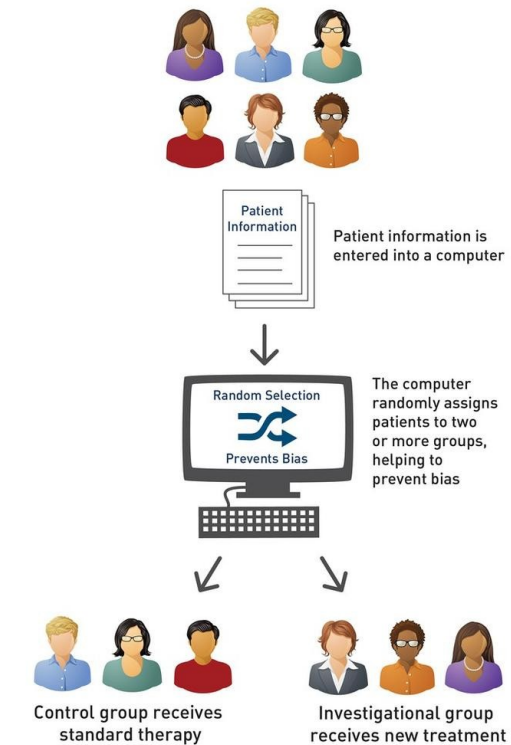
ThePrint

'Covaxin is 200% safe, don't deserve backlash': Bharat Biotech MD on questions over efficacy

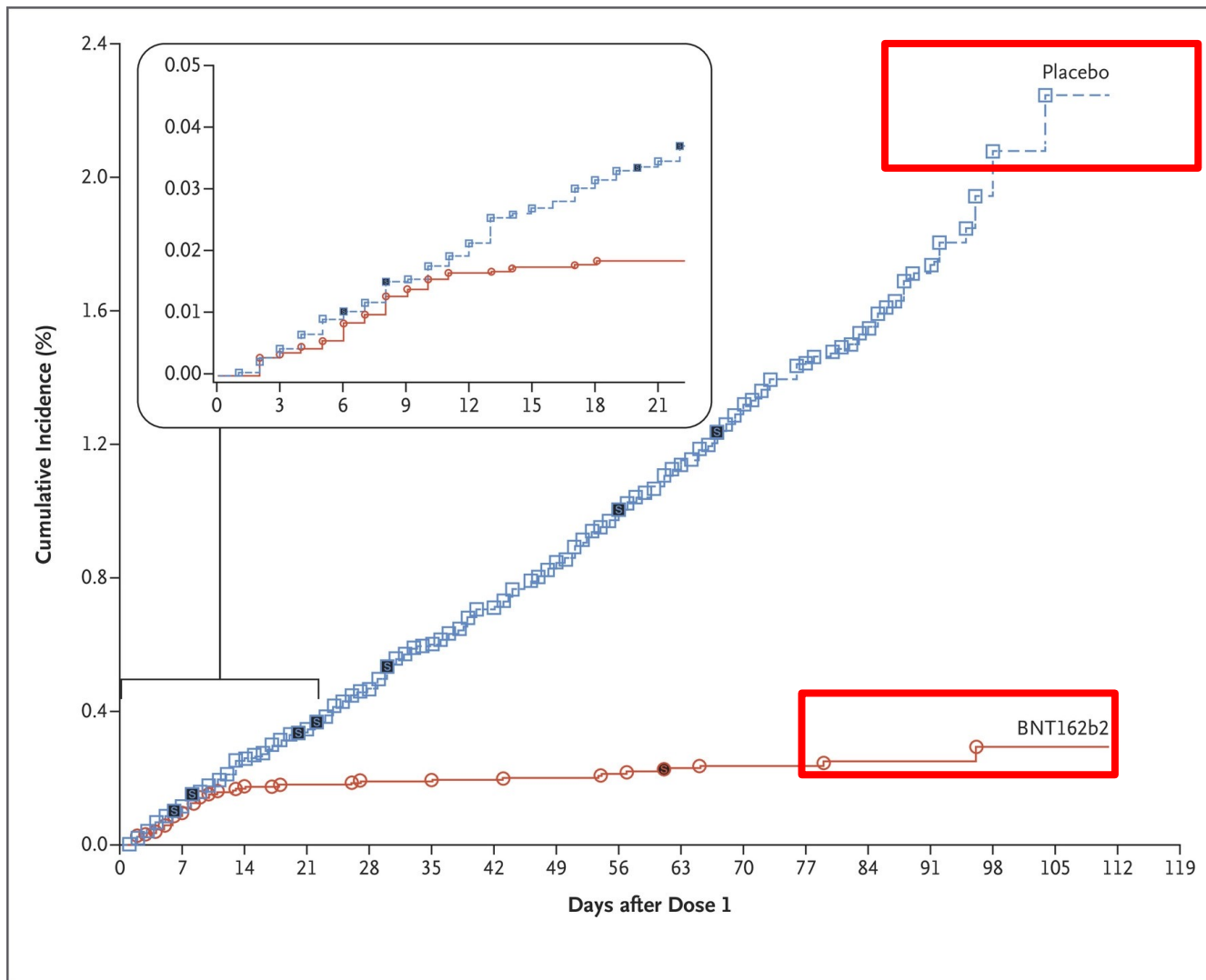
Some experts and Opposition leaders have raised concerns over the grant of approval to Covaxin, without its efficacy trials being over.

 *The Indian* **EXPRESS**

CLINICAL TRIALS RANDOMIZATION



<https://www.cancer.gov/about-cancer/treatment/clinical-trials/what-are-trials/randomization/clinical-trial-randomization-infographic>



<https://www.technologyreview.com/2020/12/10/1013914/pfizer-biontech-vaccine-chart-covid-19/>

**Data from Pfizer's
vaccine trial**

SCIENCE, NONSENSE, AND PSEUDO SCIENCE

Science refers to a systematic and organized body of knowledge in any area of inquiry that is acquired using “the scientific method”

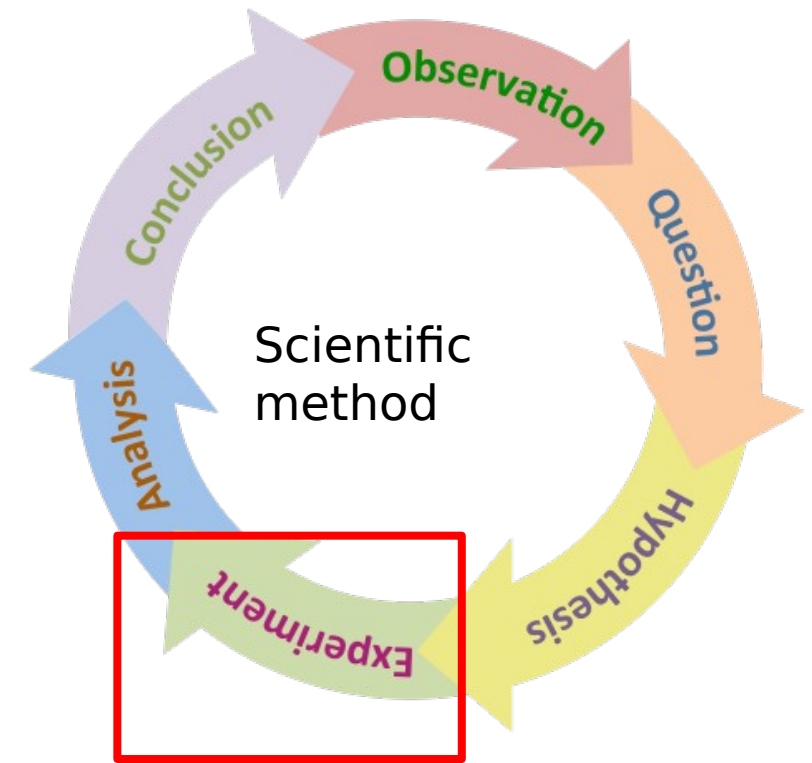


**Experimentation/
Observation
Logic and Evidence**

SCIENCE AND SCIENTIFIC METHOD

Although instances of scientific progress have been documented over many centuries, the terms **“science,” “scientists,” and the “scientific method”** were coined only in the 19th century.

Prior to this time, **science was viewed as a part of philosophy, and coexisted with other branches of philosophy such as logic, metaphysics, ethics, and aesthetics**, although the boundaries between some of these branches were blurred.



Can you experiment in social science?
What will you do when we can't

NATURAL SCIENCE AND SOCIAL SCIENCE

Natural science is the science of naturally occurring objects or phenomena, such as light, objects, matter, earth, celestial bodies, the human body, animals, environment.

Physical sciences, earth sciences, life sciences, and others...

The natural sciences are **very precise, accurate, deterministic, and independent of the person making observations**

Measuring **blood pressure**

Social science is science of **people or collections of people, such as groups, firms, societies, or economies**, and their **individual or collective behaviours**.

Psychology (the science of human behaviours)

Sociology (the science of social groups) and

Economics (the science of firms, markets, and economies).

Social sciences, which tend to be less accurate, less deterministic, less unambiguous.

Measuring
“happiness”

Social sciences tend to be less accurate, deterministic, or unambiguous.

High degree of measurement error in the social sciences and there is considerable uncertainty and little agreement on social science policy decisions.

2012

Social Science Research: Principles, Methods, and Practices

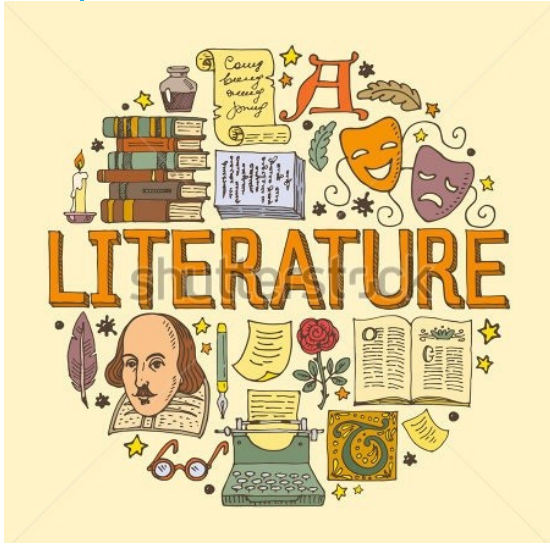
Anol Bhattacharjee

University of South Florida, abhatt@usf.edu

Introduction to Research

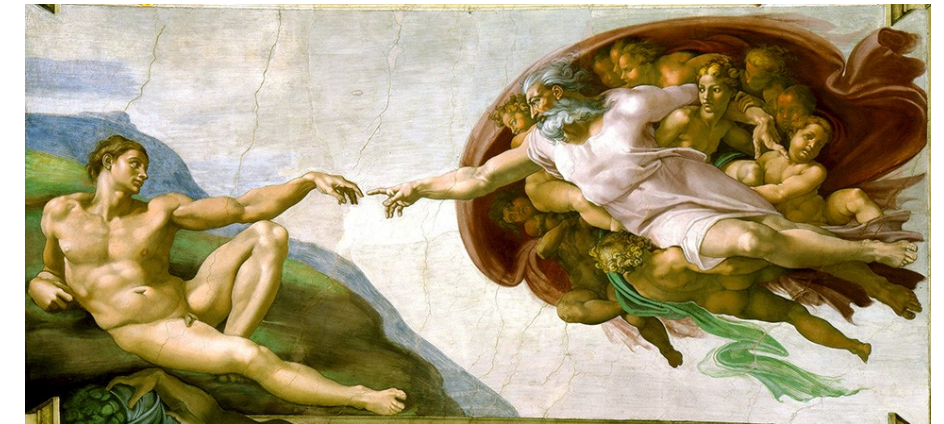
1. Science and Scientific Research....
2. Thinking Like a Researcher
3. The Research Process
4. Theories in Scientific Research.....

NON-SCIENCE IS *NOT* NONSENSE



Arts, Music, Literature, Humanities,...Law

Theology (the study of religion) is not science because theological ideas (such as the presence of God) ***cannot be tested by independent observers using a logical, confirmable, repeatable, and scrutinizable***

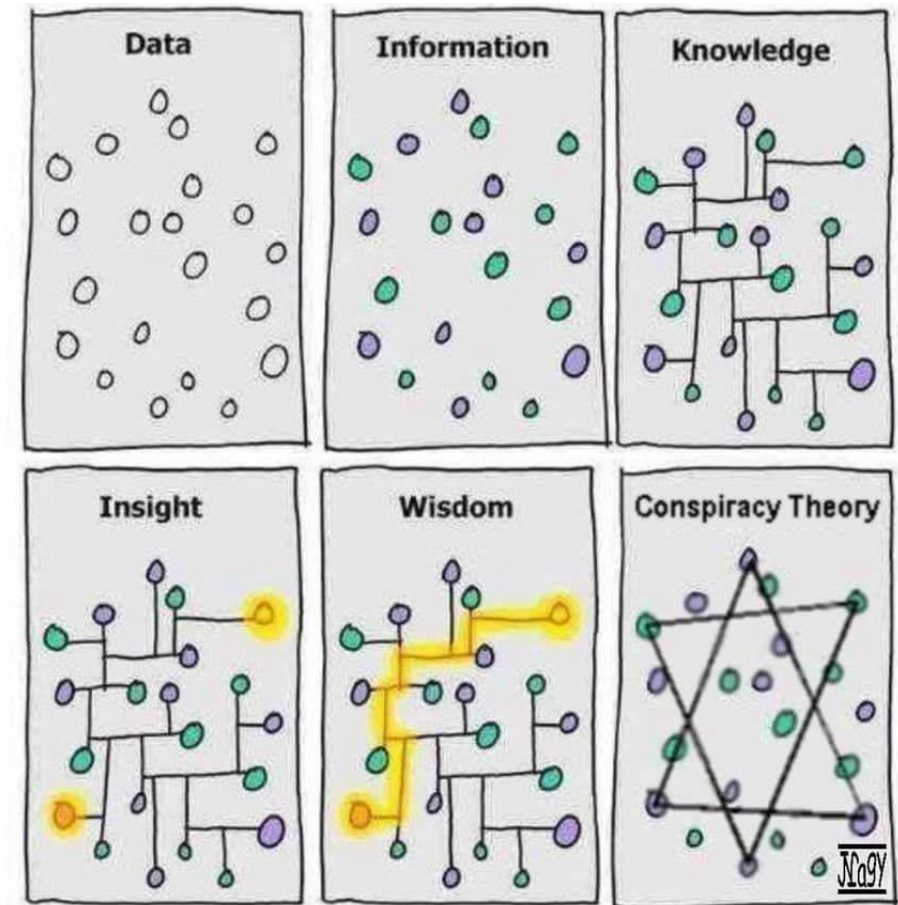


Creative and worthwhile endeavours in their own right.

SCIENTIFIC RESEARCH

Contributes to body of Knowledge

Uses “scientific method”



LAWS AND THEORIES

Scientific knowledge refers to a **generalized body of laws and theories to explain a phenomenon or behavior of interest that are acquired using the scientific method.**

Laws are observed patterns of phenomena or behaviors

Theories are systematic explanations of the underlying phenomenon or behavior.

Newton Laws, .. Boyles Law,
Laws of Thermodynamics ..
Theory of Relativity ,
Electromagnetic Theory..
Kinetic theory, ...

Are there laws in social sciences?

scientific law always applies under the same conditions, and implies that there is a causal relationship involving its elements

Theories in social sciences??

**Cognitive dissonance, Malthusian Theory,
Demographic Transition Theory,
Marxist Theory, Structural Functionalism,
Conflict theory..**

LAWS AND THEORIES

The goal of scientific research is to discover laws and postulate theories that can explain natural or social phenomena, or in other words, build scientific knowledge

The progress of science is marked by our progression over time from poorer theories to better theories, through better observations using more accurate instruments and more informed logical reasoning.

We arrive at scientific laws or theories through a process of logic and evidence.

□ **Logic (theory) and evidence (observations) are the two, and only two, pillars upon which scientific knowledge is based.**

SCIENTIFIC RESEARCH:

Scientific research operates at two levels

Theoretical level : The theoretical level is concerned with developing abstract concepts about a natural or social phenomenon and relationships between those concepts

Empirical level: concerned with testing the theoretical concepts and relationships to see how well they reflect our observations of reality, with the goal of ultimately building better theories